RESTORMER / COMBINED RESEARCH

Spatial context, gated detail correction and training-objective ablations

All eight runs continue the same trained checkpoint. Zero-start adapters preserve its initial function. Loss coefficients come from training-only gradient calibration. Raw and EMA states compete under the same validation selection rule.

Decision frozen before test: original. This is an exploratory single-seed screen; test ranks do not change the selection.

Model / experiment	Test PSNR (dB)	Test SSIM	PSNR vs original
original	32.583	0.92969	+0.000
control	32.702	0.93079	+0.119
spatial	32.708	0.93082	+0.125
detail	32.705	0.93080	+0.122
combined	32.704	0.93078	+0.120
recipe only	**32.731**	0.93166	+0.148
combined ssim	32.718	**0.93172**	+0.135
combined ffl	32.704	0.93060	+0.121
combined recipe	32.708	0.93144	+0.125

Bold identifies the highest value in a column. Headline scores weight weather, background and degradation stratum equally; snow image count does not dominate. PSNR and SSIM describe the same selected checkpoint, not separate best epochs.

WHERE THE CHANGE HELPS — AND WHERE IT DOES NOT

Experiment	Fog PSNR / SSIM	Rain PSNR / SSIM	Smoke PSNR / SSIM	Snow PSNR / SSIM
original	31.51 / 0.9666	**38.77** / **0.9768**	20.55 / 0.7937	39.51 / 0.9816
control	31.79 / 0.9682	38.70 / 0.9762	20.75 / 0.7971	39.57 / 0.9817
spatial	31.79 / 0.9682	38.71 / 0.9763	20.76 / 0.7971	39.57 / 0.9817
detail	31.79 / 0.9681	38.70 / 0.9762	20.76 / 0.7971	39.57 / 0.9818
combined	31.80 / 0.9682	38.70 / 0.9762	20.76 / 0.7971	39.57 / 0.9818
recipe only	31.80 / 0.9686	38.72 / 0.9766	**20.80** / 0.7994	39.60 / 0.9820
combined ssim	**31.81** / **0.9687**	38.71 / 0.9766	20.75 / **0.7996**	**39.60** / **0.9820**
combined ffl	31.80 / 0.9682	38.68 / 0.9759	20.78 / 0.7966	39.57 / 0.9817
combined recipe	31.79 / 0.9685	38.72 / 0.9765	20.73 / 0.7988	39.59 / 0.9819

Dataset: 18,610 training / 2,011 validation / 2,334 test pairs. Freshly decoded 52,140 active/reserve image files; no exact decoded-pair duplicates or cross-split near-copy candidates under the recorded rule. Native files and references were retained. Unknown semantic/sequence duplicates remain possible.

Smoke limitation: 31 training backgrounds and only six validation/six test backgrounds. Repeated frames do not establish independent generalization. Per-image regressions and cluster intervals are retained in test/*.json.

Interpretation: architecture-only versus control isolates added capacity; recipe-only versus control tests training changes; combined SSIM/FFL isolate auxiliaries; combined recipe tests their interaction. Raw-versus-EMA histories are available for each epoch. No Jetson latency or quantization gain is claimed.

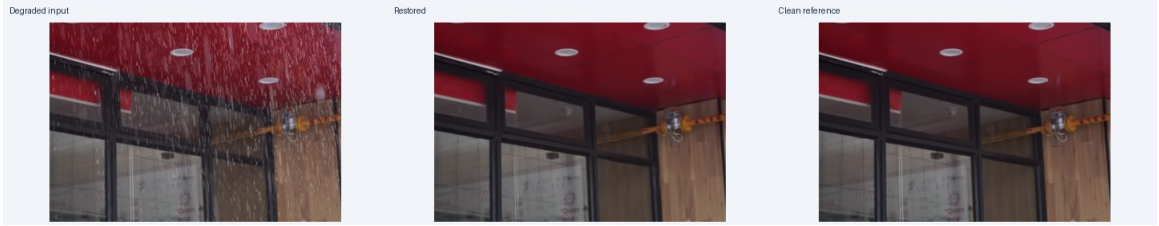
VISUAL CHECK / original / SET 1

Examples were selected by IDs before looking at quality. Each row shows degraded input, restored output and clean reference; display uses thumbnails.

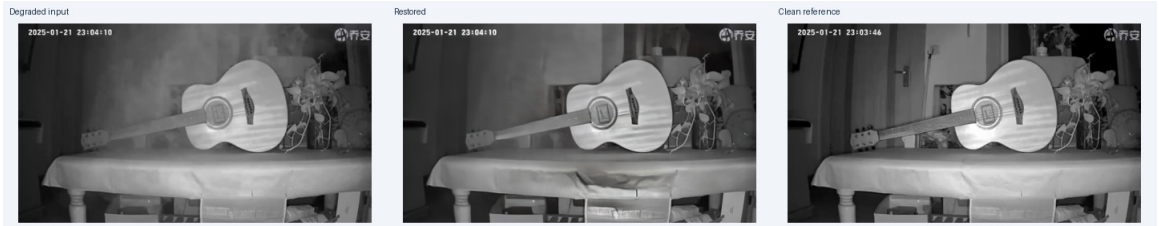
FOG · reside6k/test/0329_0.8_0.12.jpg



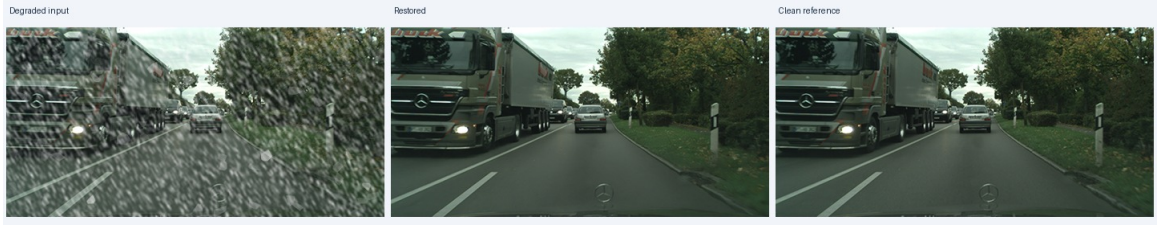
RAIN · realrain1k_h/test/767



SMOKE · smokebench_v2/train/1147



SNOW · snow_cityscapes/test/largeSnow/lindau_000001_000019_leftimg8bit



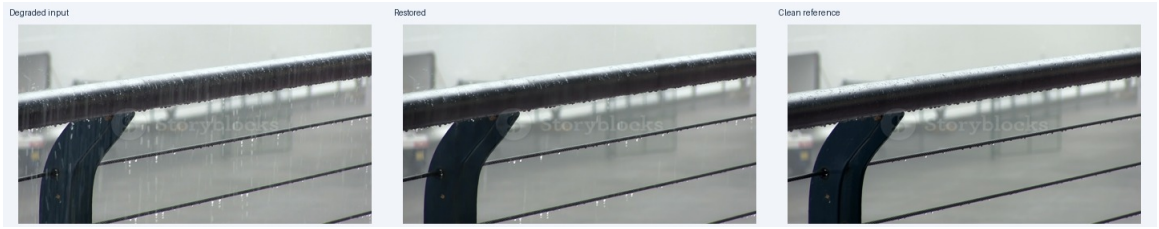
VISUAL CHECK / original / SET 2

Examples were selected by IDs before looking at quality. Each row shows degraded input, restored output and clean reference; display uses thumbnails.

FOG · reside6k/test/1006_1_0.16.jpg



RAIN · realrain1k_h/test/54



SMOKE · smokebench_v2/train/9247



SNOW · snow_cityscapes/test/mediumSnow/berlin_000153_000019_leftimg8bit

